

Domenico De Giorgio

📍 Milan, Italy | 📞 +39 389-176-6712 | ✉️ domenicodegiorgio@icloud.com
🌐 Portfolio | 🐙 DomDegi | 🌐 Domenico De Giorgio

EDUCATION

Politecnico di Milano	M.Sc. High-Performance Computing Engineering <i>Parallel Computing, Advanced Computer Architectures, Numerical Linear Algebra, Partial Differential Equations</i>	Sep 2025 – Present
Politecnico di Milano	B.Sc. Engineering of Computing Systems <i>Data Structures & Algorithms, Operating Systems, Probability & Statistics, Operations Research</i>	Sep 2022 – Sep 2025

SELECTED PROJECTS

Adaptive Space-Time Finite Element Solver	Dec 2025 – Jan 2026
- Engineered a distributed parallel solver for heat equation PDEs using the deal.II library and MPI. - Implemented space-time adaptive mesh refinement to optimize computational load across distributed nodes, significantly reducing execution time for complex finite element simulations. <i>Stack: C++, MPI, deal.II, HPC, Distributed Systems.</i>	
Advanced Monte Carlo Integration & Optimization	Dec 2025 – Jan 2026
- Built a high-performance, multithreaded library for N-dimensional Monte Carlo integration and MCMC. - Leveraged OpenMP for parallelization, heavily optimizing cache locality to maximize CPU core utilization. <i>Stack: C++, OpenMP, Stochastic Optimization, Cache Optimization.</i>	
Client-Server Multiplayer Game	Feb 2025 – Jun 2025
- Collaborated in a 4-person team to architect the complete Model and Controller layers (MVC pattern) for a real-time multiplayer digital board game. - Managed state synchronization and multi-threading through RMI and TCP/IP Sockets, achieving 97% JUnit test coverage. - Implemented a Graphical User Interface using JavaFX and SceneBuilder. <i>Stack: Java, Sockets, MVC, Maven, JUnit, JavaFX.</i>	
High-Performance CUDA Batched Matrix Multiplication	Oct 2026
- Developed a highly optimized custom CUDA kernel to accelerate batched matrix multiplications. - Maximized memory bandwidth and computational throughput by implementing shared memory tiling, thread coarsening, and bank conflict avoidance strategies. <i>Stack: C/C++, CUDA, GPU Computing, Memory Profiling.</i>	
Declarative Persistent NixOS System – Personal Daily Driver	Apr 2026 – Present
- Architected and maintained a fully reproducible Linux environment using a Btrfs subvolume layout with snapshot-based root rollback (“erase your darlings” on-disk, not in-RAM), automatically restoring the root subvolume to a clean blank snapshot on every boot. - Managed system packages and user dotfiles declaratively via Nix Flakes and Home Manager, with explicit persistence paths (via impermanence) selectively bind-mounting state that must survive reboots, enabling zero-touch provisioning and fully version-controlled system state. <i>Stack: Nix, NixOS, Btrfs, Home Manager, Linux SysAdmin, Git.</i>	

SKILLS

Languages	C, C++, Python, Java, Nix, Bash, SQL
HPC & Parallel	CUDA, MPI, OpenMP, GPU Computing, Cluster environments
Systems & Tools	Linux (Ubuntu, Arch, NixOS declarative config), Git, CMake, Slurm
CI/CD & DevOps	GitHub Actions, GitLab CI, Docker & Singularity, pytest & coverage, CodeQL
ML & Math	PyTorch, Eigen, deal.II, Numerical Analysis, Pandas, Polars
Spoken	Italian (Native), English (C1 Equivalent – TOEIC: 985/990)